



4.4.3 Via the Householder QR factorization

04.4.3 Solving LLS via Householder QR factorization

fit width



Given $A \in \mathbb{C}^{m \times n}$ with linearly independent columns, the Householder QR factorization yields n Householder transformations, $H_0, \dots, H_{n-1}$, so that

$$\underbrace{H_{n-1} \cdots H_0}_{Q^H} A = \begin{pmatrix} R_T \\ 0 \end{pmatrix}.$$

$[A, t] = \text{HouseQR_unb_var1}(A)$ overwrites A with the Householder vectors that define $H_0, \dots, H_{n-1}$ below the diagonal and R_T in the upper triangular part.

Rather than explicitly computing Q and then computing $\tilde{y} := Q^H y$, we can instead apply the Householder transformations:

$$\tilde{y} := H_{n-1} \cdots H_0 y,$$

overwriting y with $\tilde{y}$. After this, the vector y is partitioned as $y = \begin{pmatrix} y_T \\ y_B \end{pmatrix}$ and the triangular system $R_T \hat{x} = y_T$ yields the desired solution.

The steps and theirs costs of this approach are

- From [Subsection 3.3.4](#), factoring $A = QR$ via the Householder QR factorization costs, approximately, $2mn^2 - \frac{2}{3}n^3$ flops.
- From [Homework 3.3.6.1](#), applying Q as a sequence of Householder transformations costs, approximately, $4mn - 2n^2$ flops.
- Solve $R_T \hat{x} = y_T$: n^2 flops.

Total: $2mn^2 - \frac{2}{3}n^3 + 4mn - n^2 \approx 2mn^2 - \frac{2}{3}n^3$ flops.